RIG-SURJECTIVITY OF MONOMORPHISMS VIA DVR-VALUED FACTORIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5533-5543, following the statement at 5524-5531.

Editorial note. The authorial typo 'Spf(R') -> Spf(R) -> X factoring through Y' (line 5537) is preserved verbatim; the intended meaning is that the composite to Y factors through X.

Editorial note. No omissions flagged: the descent is executed via the named fpqc sheaf-condition lemma.

Allowed standard prerequisites (editorial).

- Sheaf condition for fpqc coverings. Formal Spaces, Lemma formal-spaces-lemma-sheaf-fpqc (cross-chapter, cited at ~5540-5541). X satisfies the sheaf condition for fpqc coverings; used to descend the factorization.
- Faithful flatness of complete DVR extensions. Inline standard: R/m^n -> R'/m^n R' is surjective and flat for a complete DVR extension R subset R' (used at ~5538-5539). Makes the mod-m^n diagrams an fpqc covering situation.

Source reference. Original reference 0AQD: focused-seeds/geometry/S9-restricted-monomorphism-rig-surjective/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces. Assume X and Y are locally Noetherian, f locally of finite type, and f a monomorphism. Then f is rig surjective if and only if every adic morphism $Spf(R) \to Y$ where R is a complete discrete valuation ring factors through X.*

Proof. One direction is trivial. For the other, suppose that $\mathrm{Spf}(R) \to Y$ is an adic morphism such that there exists an extension of complete discrete valuation rings $R \subset R'$ with $\mathrm{Spf}(R') \to \mathrm{Spf}(R) \to X$ factoring through Y. Then $\mathrm{Spec}(R'/\mathfrak{m}_R^n R') \to \mathrm{Spec}(R/\mathfrak{m}_R^n)$ is surjective and flat, hence the morphisms $\mathrm{Spec}(R/\mathfrak{m}_R^n) \to X$ factor through X as X satisfies the sheaf condition for fpqc coverings, see Formal Spaces, Lemma 0AQD. In other words, $\mathrm{Spf}(R) \to Y$ factors through X. □

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